Immunoprecipitation Protocol

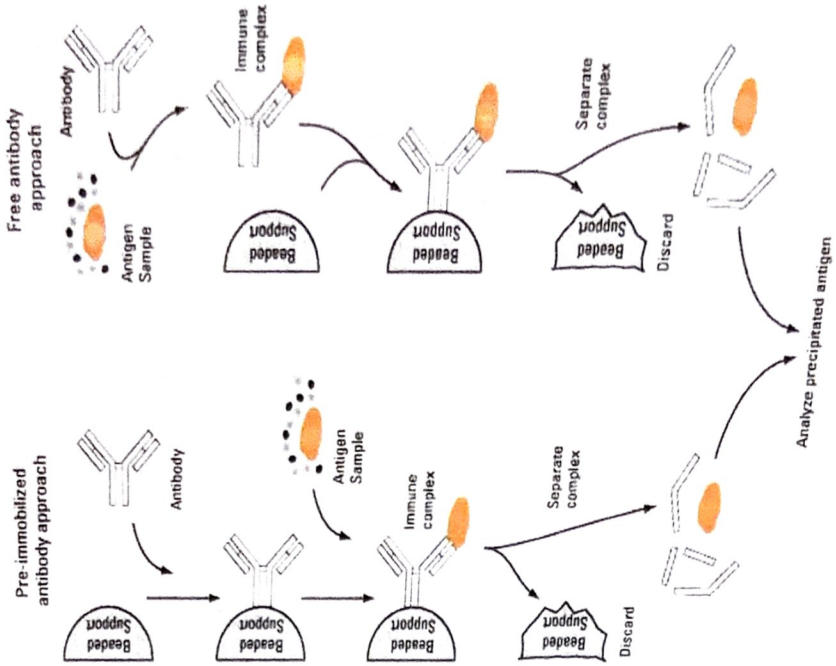
Holaska Lab, Aayush Gupta
RVSTBES – Molecular Cell Biology & Neuroscience

Safety statement: Prior to beginning an experiment, please read the Standard Operating Procedure (SOP) for this protocol as well as the Safety Data Sheets (SDS) for all chemical reagents listed. The SOP will contain a list of all chemicals for this protocol as well as directions for handling and disposal of hazardous wastes. It will also contain proper PPE guidelines and emergency protocols for accidents involving hazardous chemicals. Always be aware of your surroundings in the lab, and be sure to communicate any hazards to those around you when performing an experiment.

Hazard statement: Any hazardous chemicals handled within this protocol will be marked with the following notation: (H). Please refer to the associated SOP to better understand the hazards and how to work with and dispose of these chemicals safely.

Background:

Immunoprecipitation (IP) is the small-scale affinity purification of antigens using a specific antibody that is immobilized to a solid support such as magnetic particles or agarose resin. Immunoprecipitation is one of the most widely used methods for isolation of proteins and other biomolecules from cell or tissue lysates for the purpose of subsequent detection by western blotting and other assay techniques. Immunoprecipitation was first developed as an adaptation of traditional column affinity chromatography, which involves allowing sample, wash, and other solutions to pass through a column that is packed with porous resin (typically beaded agarose) onto which a target-specific antibody has been immobilized. However, instead of using a packed column, immunoprecipitation uses a small amount of resin in a microcentrifuge tube, and incubation steps are performed in a batch-wise manner. Unlike column affinity chromatography, the goal of immunoprecipitation is to isolate just enough protein to be able to measure it by western blotting or other semi-quantitative or quantitative assay methods (ThermoFisher Scientific).



Equipment:

1. Magnetic tube holder for 1.5mL Eppendorf tubes
2. Rotating tube holder for 1.5mL Eppendorf tubes
3. Sonicator
4. Centrifuge
5. Vortex

Reagents:

1. 1X phosphate-buffered saline (PBS)
2. Dynabeads™ Protein G Immunoprecipitation Kit (Invitrogen #10007D)
    a. Dynabeads Protein G magnetic beads
    b. Ab-binding & wash buffer
    c. Elution buffer
    d. Wash buffer
3. Modified NEHN buffer in deionized water:
    a. 500mM NaCl
    b. 20mM HEPES, pH 8
    c. 20% glycerol
    d. 1% NP-40
    e. 1mM EDTA
    f. 2mM DTT
    g. 1mM PMSF
    h. 5µg/mL each aprotinin, leupeptin, pepstatin A
4. NEHN Dilution buffer in deionized water:
    a. 20mM HEPES
    b. 20% glycerol

c. 1mM EDTA
d. 2mM DTT
e. 5µg/mL each aprotinin, leupeptin, pepstatin A

5. Wash buffer in deionized water:
   a. 150mM NaCl
   b. 20mM HEPES
   c. 0.2mM EDTA
   d. 0.5mM DTT
   e. 20% glycerol
   f. 0.2% NP-40
   g. 5µg/mL each aprotinin, leupeptin, pepstatin A
6. For free antibody approach, use IP buffer in deionized water:
   a. 50mM HEPES, pH 8
   b. 300mM NaCl
   c. 0.3% Triton X-100
   d. 1mM PMSF
   e. 1X Halt™ protease inhibitor cocktail
7. 2X Laemmli sample buffer supplemented with 100mM DTT

Procedures:

Pre-immobilized antibody approach

Formation of the immune complex:

1. Label 1.5mL Eppendorf tubes for each sample and pipette 200µL 1X PBS into each tube.
2. Gently vortex magnetic beads and pipette appropriate volume into each tube.
   a. 30µL magnetic beads will isolate approximately 5-6µg IgG.
3. Place on rotator to wash the magnetic beads (2 min/wash).
4. Snap-spin tubes to remove any liquid trapped in the caps, then place tubes on magnet for 1 min.
5. Remove supernatant and repeat wash steps 2X more.
6. During third wash, prepare antibody solution:
   a. 2-10µg antibody per 500µg protein lysate (~200-300µg protein in $1x10^6$ cells).
   b. Total volume should be 200µL.
7. Incubate at 4°C on a rotator for 2 hours.

Lysis of cells and solubilization of proteins:

1. Harvest cells by trypsinization and pellet by centrifugation at 300xg for 5 min.
2. Remove supernatant and discard. Resuspend cell pellet in appropriate volume of PBS and count using the Countess automated cell counter.
3. Pellet cells at 300xg for 5 min and remove supernatant. Snap freeze in liquid nitrogen and store at -80°C if not using immediately. Can be stored for 1 month.
4. Lyse cell pellet with ice-cold modified **NEHN** buffer: 100µL/$1x10^6$ cells.
   a. Ensure that you pipette up and down to break up any clumps that are visible.

b. If precipitating nuclear proteins, you may need to add benzonase to the lysis buffer at a final concentration of 50 units/mL of sample. This will shear up the chromatin better than sonicating alone, thus freeing more of the proteins into the soluble fraction.

5. Incubate on ice for 10 min, occasionally inverting the tubes to mix.
6. Sonicate sample at 40% amplitude for 20 rounds of 5 sec on, 10 sec off, performed on ice.
    a. Make sure the sample does not froth, otherwise the sample **cannot** be used.
    b. If using a probe sonicator, position the probe so that it is fully submerged in the suspension but does not touch the walls of the tube.
    c. Place the tube containing cell lysate in a pre-chilled metal tube holder on ice so that the sample remains ice cold.
    d. After positioning the sonicator in the suspension, test whether it is in contact with the walls of the tube by flicking the lid of the Eppendorf tube. If the tube spins freely with no resistance, it is most likely in the correct position for sonication.
7. Centrifuge sample at 20,000xg for 20 min at 4°C.

Immunoprecipitation:

1. Retrieve antibody-bound magnetic beads from rotator and place on magnet.
2. Remove supernatant and discard, then wash beads twice with 1X PBS and once with NEHN **dilution** buffer.
3. Retrieve sample from the centrifuge and carefully transfer the soluble lysate (supernatant) to a new tube.
4. Dilute soluble lysate 1:1 with NEHN dilution buffer, then add directly to Ab-bound magnetic beads, distributing the volume equally amongst the samples.
    a. Remember to take input sample BEFORE placing on magnetic beads.
5. Rotate at 4°C for 2 hours or overnight.
6. Snap-spin tubes and place on magnet. Discard supernatant.
    a. Remember to take unbound sample.
7. Wash beads 5X with modified wash buffer.
    a. If testing an antibody for ability to immunoprecipitate, keep washes to make gel samples.
8. Resuspend each sample in 50µL 2X LDS sample buffer and heat on a Thermomixer for 15 min at 75°C, shaking at 1000rpm.
9. Snap-spin tubes, place on magnet for 1 minute, and load supernatants on SDS-polyacrylamide gel (see Western blotting protocol).

Free antibody approach

Lysis of cells and solubilization of proteins:

1. Harvest cells by trypsinization and centrifuge at 300xg for 5 min.
2. Remove supernatant and discard. Resuspend cell pellet in appropriate volume of PBS and count using the Countess automated cell counter.

3. Pellet cells at 300xg for 5 min and remove supernatant. Snap freeze in liquid nitrogen and store at -80°C if not using immediately. Can be stored for 1 month.
4. Lyse cell pellet with ice-cold **IP buffer**:100µL/1x10^6 cells.
5. Vortex for 15 seconds, then place on ice for 1 minute. Repeat this 5 times to break up DNA. Consider adding benzonase at 50 units/mL of sample if proteins are not adequately solubilized
6. Centrifuge sample at 20,000xg for 20 min at 4°C.

Immunoprecipitation:

1. Remove samples from the centrifuge and transfer supernatant to a new tube.
    a. Wash the insoluble fraction twice with 1X PBS and spin at the same speed used for separating the fractions. Resuspend pellet in appropriate volume of PBS and mix gel sample with equivalent volume of 2X Laemmli sample buffer.
2. Add precipitating antibody to the soluble lysate at an appropriate concentration and incubate at 4°C for 2 hours with end-over-end rotation.
    a. All new antibodies used for IP should be optimized for precipitating at least 90% of antigen from the soluble lysate.
3. Wash Dynabeads protein G magnetic beads 3X with 200µL **IP buffer** (3min/wash).
4. Add beads to the sample and incubate at 4°C for 2 hours with end-over-end rotation. Stop
5. Snap-spin tubes and place beads on magnet for 1 minute. Remove a small aliquot for gel sample. This is the unbound fraction. Discard the rest of the supernatant and wash the beads 5X with "Wash buffer" from the Dynabeads Immunoprecipitation kit.
6. Elute protein-antibody complexes from the beads by resuspending in 2X Laemmli sample buffer and heating for 15 minutes at 75°C.
7. Snap-spin tubes, place on magnet for 1 minute, and load supernatant on gel.

...noPercipitation Protocol (IP) Notes

...amentally isolating specific proteins from a cell lysate (sup)
...d on binding the antibody and antigen, and pulling that away from everything else.

...cifically, we use Protein G magnetic beads, which bind to the
...of antibodies, also known as **Constant** region.
...d - Protein G - Antibody - Protein - Secondary Protein (Chain)

700 ug/ml → 2.85 ml
~~.5 ug/ml~~
500 ug/ml 4 ml

...-immobalized Approach
...o stick the antibodies to the beads first, then pour cell lysate on top
...ur off excess supernatent, wash away debris, and boil beads for pure protein.

...ce-Antibody Approach
You first let antibody float and attach in cell lysate first
You then let the magnetic beads in to attach to constant region
You then use a magnet to pull away entire complex, wash it down.

...used for Co-IP, where a primary protein is pulled down along with
...r secondary protein.

Immobalized Approach Steps
...abel tubes with each sample and add 200 ul of 1X PBS to each tube
Vortex the magnetic beads, then add appropriate amount to tubes
Place on rotator to wash for 2 minutes. Snap-spin tube.
Put tube on magnet for 1 minute, then remove liquid buffer. Repeat twice.
Prepare 200 ul antibody solution, incubate tube at 4°C for 2 hours.

(Bead Attachment to Tubes)

...t 2: Lysis of Cells and Solubilization of Proteins
Add trypsin to deattach cells, spin at 300 xg for 5 minutes.
Discard excess media from pellet and rinse in PBS.
Count cells, spin them down again, flash freeze at -80°C if not used immediately.

Preparation of Cells

Then add ice-cold modified NEHN lysis buffer for 100 ul for each 1 million of cells.
Pipette vigorously to break up cell chunks and dissolve cell membranes.
- Can also add 50 ul/ml of benzonase to release proteins **inside Nucleus**
- Let sample sit on ice for 10 minutes, occassionally mixing, sonicate at 40% for 20 rounds, 5 on and 10 off
- Centrifuge sample at 20,000 xg for 20 minutes at 4°C.

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col Notes Continued

Immunoprecipitation (Capture Stage)

beads off rotater, put on magnet, wash 2x with PBS, once with NEHN
tube out, pipette clear liquid into new tube, add 1:1 NEHN buffer. Pour onto beads directly.
putting on beads, save small sample as "input" sample to prove protein existed in first place.
mixture on rotator for 2 hours at 4°C.
tubes spin, place on magnet, pull off liquid. Save some liquid to prove all protein was attached & taken.
5 times with modified wash buffer, add buffer, remove excess liquid, remove all unnecessary junk.
50 µl of heavy 2x LDS Buffer and cook on shaker for 15 mins at 75°C.
take on magnet, remove liquid which is all concentrated protein.

Immunoprecipitation Protocol (HELA & C2C12 Cell Lines)

Retrieve both samples from -80°C freezer.

→ Estimating around 2ml

→ Create IP Buffer 100µl / 1×10^6 cells, EXACT amount of IP:] Buffer Formula

→ Lyse the cell pellet with created IP Buffer and leave on ice for 10 mins.

→ Add 50 units of benzonase to each sample to enhance breakdown
Centrifuge at 20,000 xg for 20 minutes at 4°C

Immunoprecipitation

- Wash insoluble fraction with PBS (→ supernatant) and also centrifuge at 20,000 xg for 20 mins] → Before this transfer supernatant to new tubes
  - → Resuspend in PBS and add equivalent volume of 2x Laemmli Buffer
  - → Add antibodies to soluble lysate at 10µl / 1µg. Incubate on rotation for 2 hours at 4°C.
  - → Wash beads 3 times each with 200 µl of IP Buffer. Spin for 1 min in between.

Add beads to the sample, incubate on rotation at 4°C for 2 hours.

Where and what tubes are labeled for each sample???

→ Total Cell Lysate, Insoluble Fraction, Soluble Fraction, & Input Tubes for each.